

The background is a blurred image of a computer monitor. The screen shows a dark-themed code editor with some text in blue and green. At the bottom, a Windows taskbar is visible with various application icons like Chrome, Photoshop, and others. The text is overlaid on this background.

Introduction To Programming Using C++ Level 2

Update Record In File



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Definition

 `UpdateRecordInFile()`

 `UpdateRecordInFile()` is a function used to find a specific record in a text file and replace it with a new value.

 The remaining records are preserved.

Explanation

Updating File Data

-  The same general idea used for deleting a record is used here.

-  We load the file contents into a `vector<string>`, modify the required element in memory, and then save the vector back to the file.

`PrintFileContent()`

-  Displays the file contents before and after the update.

`LoadDataFromFileToVector()`

-  Reads every line from the file and stores it in the vector.

`SaveVectorToFile()`

-  Writes the modified vector back to the file using `ios::out`.

`UpdateRecordInFile()`

-  Searches for the requested record.

-  When the record is found, it is replaced with the new value.

Procedures

1. Load the file contents into a `vector<string>`.
2. Loop through every element in the vector.
3. Compare each element with the record we want to update.
4. If a match is found, replace it with the new value.
5. Save the modified vector back to the file.
6. Print the file contents again to verify the update.

💡 Example

```
void UpdateRecordInFile(string FileName, string Record, string UpdateTo)
{
    vector<string> vFileContent;

    LoadDataFromFileToVector(FileName, vFileContent);

    for (string& Line : vFileContent)
    {
        if (Line == Record)
        {
            Line = UpdateTo;
        }
    }
    SaveVectorToFile(FileName, vFileContent);
}
```

💡 Example

```
int main()
{
    cout << "File Content Before Update:\n";
    PrintFileContent("MyFile.txt");
    UpdateRecordInFile("MyFile.txt", "Ali", "Omar");
    cout << "\n\nFile Content After Update:\n";
    PrintFileContent("MyFile.txt");
    return 0;
}
```

➡ Result

Before update:

Ali

Shadi

Maher

Fadi

Lama

After updating `Ali` to `Omar`:

Omar

Shadi

Maher

Fadi

Lama

Note

- ◆ The same general logic used for deletion is used here, but instead of setting the element to an empty string, we assign the new value.
- ◆ The update happens inside the vector in memory.
- ◆ After the modification, the complete vector is saved back to the file.
- ◆ If the requested record does not exist, no element is changed.

⊘ Warning

- ◆ If the same record appears multiple times, this function will update all matching records.
- ◆ Make sure ``UpdateTo`` contains the intended new value before saving the vector.
- ◆ ``ios::out`` rewrites the file contents, so the vector should contain all records that need to be preserved.



Comparison

Delete Record

- ✓ `Line = ""`
- ✓ Removes the matching record from the resulting file.
- ✓ Used when the record is no longer needed.

Update Record

- ✓ `Line = UpdateTo`
- ✓ Replaces the matching record with a new value.
- ✓ Used when the record should remain but its value needs to change.

✦ Summary

- ✓ Load the file contents into a vector.
- ✓ Search for the required record.
- ✓ Replace the matching record with the new value.
- ✓ Save the modified vector back to the file.
- ✓ If no matching record exists, nothing is changed.

Thank you 😊